FINAL PROJECT REPORT

Global Fuel Prices and Taxation Trends

A Comprehensive Analysis using Tableau

Project details	Information
Project title	Global Fuel Prices and Taxation Trends: A Comprehensive Analysis
Project area	Data Analytics and Visualization
Primary tool	Tableau / Tableau Public
Supporting website	Flask, HTML, CSS and Bootstrap
Dataset	global_fuel_prices_Dataset.csv
Project period represented	Dataset dates: 6 January 2020 – 6 April 2026
Report date	27 October 2026
Submission deadline	28 October 2026
Team ID	To be added
Team member names	Krishnakant Rathi

Prepared for academic project submission. This report documents the described implementation and verified dataset profile. Any item marked “to be added” requires team confirmation before final submission.

Document Control and Executive Summary

This reportconsolidates the project's problemdefinition, userperspective, requirements, design, planning, implementation, testing approach, results, limitations and future scope. The project uses a structured fuel-price dataset and Tableau visualizations to make cross-country and time-based comparisons easier to interpret. A supporting Flask website provides a project presentation layer.

Item	Details
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Analysis
Version	Final report draft for review
Prepared on	19 October 2026
Team ID	To be added
Dashboard	Open published Tableau Dashboard
Stories	Global Trend Story Fuel Price comparision across Top 3 countries Taxation Country and regional differences
Source repository	GitHub repository

Executive Summary

Fuel-price data is difficult to compare when records span multiple countries, dates, fuel types, taxation levels, subsidy categories and crude-oil prices. This project organizes these variables into Tableau worksheets, a consolidated dashboard and a Tableau Story. The dashboard supports exploration; the Story arranges the analysis into a guided sequence. The supporting website introduces the project and can link viewers to published outputs.

The dataset contains 27,468 rows and 10 columns for 84 countries, with dates from 6 January 2020 to 6 April 2026. A local validation check found no missing values and no exact duplicate rows. The report's observations are descriptive: visual associations do not establish causation.

Submission note: add the public dataset source URL and the final demo-video URL in the appendix before submitting. The local CSV filename is known, but its public download/source URL has not been provided.

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Abbreviations

Term	Meaning
CSV	Comma-Separated Values file format
LPG	Liquefied Petroleum Gas
KPI	Key Performance Indicator
UI	User Interface
DFD	Data Flow Diagram
QA	Quality Assurance

1. Introduction

1.1 Project Overview

The project analyzes global petrol, diesel and LPG prices alongside Brent crude-oil prices, tax percentage, subsidy level, region and income level. Tableau is used to create country comparisons, trend charts and related views, then combine them into a dashboard and a Story for interpretation. A Flask-based website built with HTML, CSS and Bootstrap serves as a supporting project page.

1.2 Purpose

The purpose is to transform a large tabular dataset into understandable visual evidence. The dashboard is intended to help users explore price differences and time trends without manually scanning thousands of records. The Story provides a logical path through the major questions.

1.3 Objectives

- Summarize fuel-price patterns across countries and dates.
- Compare petrol, diesel and LPG using consistent labels and units.
- Explore tax percentage and subsidy-level differences in the dataset.
- Compare fuel-price trends with the Brent crude-oil series as context.
- Examine differences by region and income-level groups.
- Publish the Tableau dashboard and Story and document the outputs.

1.4 Scope and boundaries

Included: dataset review, field inspection, comparative and time-series visualizations, dashboard and Story design, project website presentation, documentation and performance-check planning. Excluded: forecasting, causal inference, live fuel-price feeds, a separate Snowflake warehouse and claims about policy impact that cannot be supported by the dataset.

1.5 Intended users

Potential users include students, data analysts, researchers and readers interested in fuel markets, energy costs and public policy. The project is an educational descriptive-analytics solution, not a real-time price service or policy recommendation engine.

2. Ideation Phase

2.1 Problem Statement

Fuel-price information is distributed across countries, dates and fuel types. When taxation, subsidy levels and crude-oil prices are reviewed separately, it can be difficult to compare patterns consistently. The project addresses this problem by consolidating key variables into Tableau views and a guided Story.

2.2 Empathy Map Canvas

Dimension	User perspective
Says	“I need to compare fuel prices across countries.” “How do taxes and subsidies relate to the prices shown?”
Thinks	The raw dataset has many fields; a clear view may help identify which comparisons are useful.
Does	Compares countries, fuel types and dates; reviews tax/subsidy fields and crude-oil trends.
Feels	May feel overwhelmed by large tables and uncertain about which patterns matter; clearer charts can improve confidence.
Persona	Student, analyst or researcher exploring global fuel-price data.

2.3 Brainstorming and prioritization

Ideas considered for the solution included country-level fuel-price comparisons, time-series analysis, fuel-type comparisons, tax/subsidy analysis, and comparison with Brent crude oil. The chosen direction combines these analyses in one Tableau dashboard and organizes the findings in a Story.

Idea	Value to user	Priority
Compare petrol, diesel and LPG across countries	Enables country and fuel-type comparisons	High
Review fuel-price trends over time	Shows changes and broad patterns	High
Compare tax and subsidy categories with prices	Adds policy-related dimensions for exploration	High
Compare fuel prices with Brent crude oil	Adds contextual trend comparison	High
Analyze by region and income level	Supports group-level comparison	Medium–High

Selection rationale: the selected solution makes the strongest use of the available dataset fields while keeping the project within the scope of descriptive visualization.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User action	Challenge	Project response
Discover	Finds a question about global fuel prices	Information is spread across fields and countries	Introduce the dataset and main questions
Explore	Compares countries, fuel types and dates	Manual table comparisons are time-consuming	Use Tableau charts and comparative views
Interpret	Reviews tax, subsidy, region, income and crude-oil context	Relationships may be misread as causal	Provide labels and a guided Story
Share	Opens dashboard/Story and discusses findings	Need a simple way to access outputs	Publish links and include them in the website/report

3.2 Functional Requirements

- Load the CSV dataset into Tableau and use the available fields.
- Compare average petrol prices across countries.
- Display global fuel-price trends over time.
- Compare petrol, diesel and LPG prices.
- Visualize tax percentage and subsidy-level differences.
- Compare patterns by income level and region.
- Display Brent crude-oil trends alongside fuel-price trends.
- Combine selected views into a dashboard and create a Tableau Story.
- Publish dashboard and Story links; provide a supporting project website.

3.3 Non-functional Requirements

- Readability: chart titles, legends, units and labels should be understandable.
- Usability: viewers should be able to navigate the dashboard and Story.
- Reliability: visuals should render and links should open as expected.
- Performance: dashboard loading and filtering should be checked on the intended device/network.
- Maintainability: project files and documentation should be organized.
- Interpretation: claims should remain within the dataset's coverage and not imply causation without appropriate analysis.

3.4 Data Flow Diagram — textual summary

SourceCSV → Tableau dataconnection → field/type and data-quality review → worksheets and comparisons → consolidated dashboard → Tableau Story → Tableau Public. Separately, the Flask website presents project information and links to the published analytical outputs. No Snowflake database is part of the described implementation.

3.5 Technology Stack

Component	Technology	Role
Source data	CSV file	Stores the fuel-price records and dimensions.
Analytics and visualization	Tableau Desktop / Tableau Public	Creates worksheets, dashboard and Story.
Website backend	Python and Flask	Serves the project website and its routes.
Website UI	HTML, CSS, Bootstrap	Provides layout and responsive presentation.
Publishing and version control	Tableau Public, GitHub where files are committed	Shares analytics and project source files.

4. Project Design

4.1 Problem–Solution Fit

The user need is to compare fuel prices across countries and time while viewing taxation, subsidy levels, income/region groups and Brent crude oil in one analytical context. The Tableau dashboard and Story address this need by combining comparable visualizations and guiding the viewer from broad trends to more specific comparisons. The solution supports exploration, not proof of causal relationships.

4.2 Proposed Solution

Prepare and inspect the CSV, connect it to Tableau, create views for the main business questions, arrange selected views in a dashboard, sequence the analysis in a Story, publish the outputs, and present the project through a Flask/Bootstrap website.

4.3 Solution Architecture

Layer	Input / component	Output
Data source	global_fuel_prices_Dataset.csv	Structured records with date, country, region, income, subsidy, fuel prices, crude-oil price and tax percentage.
Preparation	Tableau data connection and field review	Usable fields and consistent analytical dimensions.
Analysis	Tableau worksheets	Country comparisons, fuel-type views, trends, tax/subsidy analysis and group comparisons.
Presentation	Tableau dashboard	Consolidated interactive exploration.
Storytelling	Tableau Story	Ordered sequence of views and explanatory observations.
Website	Flask, HTML, CSS, Bootstrap	Project overview and links to published outputs.

4.4 Key design decisions

- Use country-level bars for comparing values and line charts for time-series trends.
- Separate the dashboard from the Story: the dashboard supports exploration; the Story supports a guided narrative.
- Use available dataset dimensions for filtering and segmentation where implemented.
- Avoid presenting correlation or visual co-movement as proof that one factor causes another.
- Do not add Snowflake to the architecture because it was not used in the described project.

5. Project Planning and Scheduling

The planning documentuses5 October2026as the startofthe projectvalidation/progress-check period and 25 October 2026 as the submission deadline. Dates are planning milestones; actual completion should be marked only after verification.

Period	Planned activity	Evidence / output
5–9 October	Dataset validation and core country/fuel comparisons	Dataset profile and core worksheets
10–14 October	Trends, tax, subsidy, crude oil and group analysis	Additional worksheets and comparison views
15–18 October	Dashboard, Story and website review	Published dashboard/Story and website links
19–21 October	End-to-end testing and issue correction	Completed testing checklist and corrected links
22–26 October	Documentation, screenshots and demo preparation	Final report, template PDFs, screenshots and demo plan
27 October	Final review and submission	Submission folders and final video/link

5.1 Work breakdown

- Dataset and field validation.
- Business question and visualization mapping.
- Tableau worksheets, dashboard and Story.
- Flask website presentation and published links.
- Performance checks, evidence capture and documentation.
- Final PDF review, demo rehearsal and submission.

5.2 Planning assumptions

The sprint points and schedule are planning estimates, not independently measured effort. Team member names and Team ID were not provided and must be filled by the team. Do not represent planned tasks as completed unless they have been checked.

6. Functional and Performance Testing

6.1 Functional testing checklist

Test ID	Test case	Expected result	Status
FT-01	Open source CSV and confirm expected fields	10 expected columns are available	Needs final team verification
FT-02	Review date range and row count	Dataset profile matches report values	Validated locally
FT-03	Open each main Tableau view	Charts render and show meaningful labels	Needs final team verification
FT-04	Compare fuel types and country values	Views use the intended fields and aggregation	Needs final team verification
FT-05	Check tax/subsidy and group comparisons	Categories and axes are readable	Needs final team verification
FT-06	Open published Dashboard URL	Dashboard page loads	Verify on submission device
FT-07	Open published Story URL	Story points display in sequence	Verify on submission device
FT-08	Open Flask website and links	Page and navigation links work	Needs final team verification
FT-09	Review text, units and interpretation	No unsupported causal claims or ambiguous units	Needs final team verification

6.2 Dataset quality checks (verified locally)

Check	Observed result
Record count	27,468 rows
Column count	10 columns
Country coverage	84 unique countries
Date range	2020-01-06 to 2026-04-06
Missing values	0 across the 10 columns
Exact duplicate rows	0
Regions	7 unique region labels
Income levels	3 unique categories
Subsidy levels	4 unique categories

6.3 Performance testing approach

The performance template covers data rendered, data preprocessing, filter utilization, calculated fields, dashboard design and Story design. This report does not invent response times, load times, browser measurements or filter latency. Capture those values by opening the published Tableau outputs on the intended device and recording the actual results.

- Open the dashboard and Story; record actual load times on the device/browser used for submission.
- Test available filters and controls; confirm views update and labels/legends remain readable.
- Count actual visualizations and capture screenshots of successful rendering or any issues found.

7. Results and Output Evidence

7.1 Dataset summary

The dataset contains 27,468 records and 10 columns for 84 countries, covering 2020-01-06 through 2026-04-06. It includes date, country, region, income level, subsidy level, petrol price, diesel price, LPG price, Brent crude price and tax percentage. Local validation found 0 missing values and 0 exact duplicate rows.

Measure	Mean	Minimum	Maximum
Petrol (USD/litre)	2.282	0.010	6.779
Diesel (USD/litre)	2.134	0.010	6.240
LPG (USD/litre)	1.711	0.010	5.100
Brent crude (dataset units)	107.982	73.735	130.000
Tax percentage	26.77	0.00	65.00

These figures are computed from the local CSV file. The dataset's value ranges may include extreme or synthetic values; interpret averages and minimum/maximum values in context. Units for Brent crude should be confirmed against the dataset documentation before final submission.

7.2 Tableau outputs

Output / view	Purpose
Average Petrol Price by Country	Compare petrol price levels across countries.
Global Fuel Price Trends	Inspect broad movement over the date range.
Fuel Type Comparison	Compare petrol, diesel and LPG values.
Tax Percentage by Country	Review reported tax percentage differences.
Subsidy Analysis	Compare price patterns across subsidy categories.
Price by Income Level	Compare price patterns by income group.
Diesel and LPG country views	Inspect country differences for diesel and LPG.
Brent Crude Oil Trend	Review crude-oil movement and compare visually with fuel trends.

7.3 Published output links

Dashboard: Open published Tableau Dashboard

Stories: Global Trend Story

Fuel Price comparision across Top 3 countries

Taxation

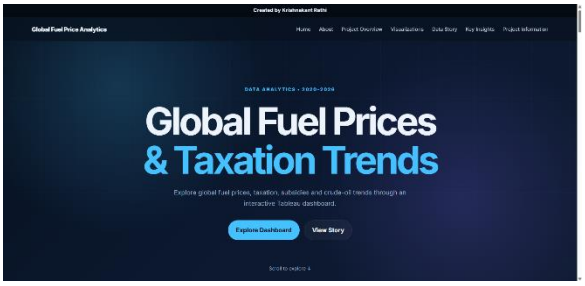
Country and regional differences

Repository: GitHub repository

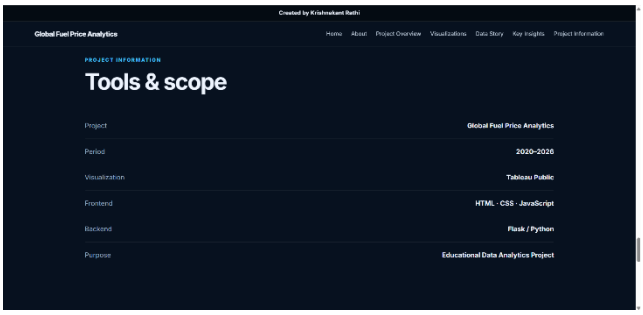
Screenshot evidence to add

- Insert a screenshot of the full Tableau Dashboard with the title and major charts visible.
- Insert a screenshot of the Tableau Story showing its first/overview point.
- Insert one or two screenshots showing representative chart details or interactive filters.
- Insert a screenshot of the project website homepage.
- If available, include evidence of the dataset file and GitHub repository.

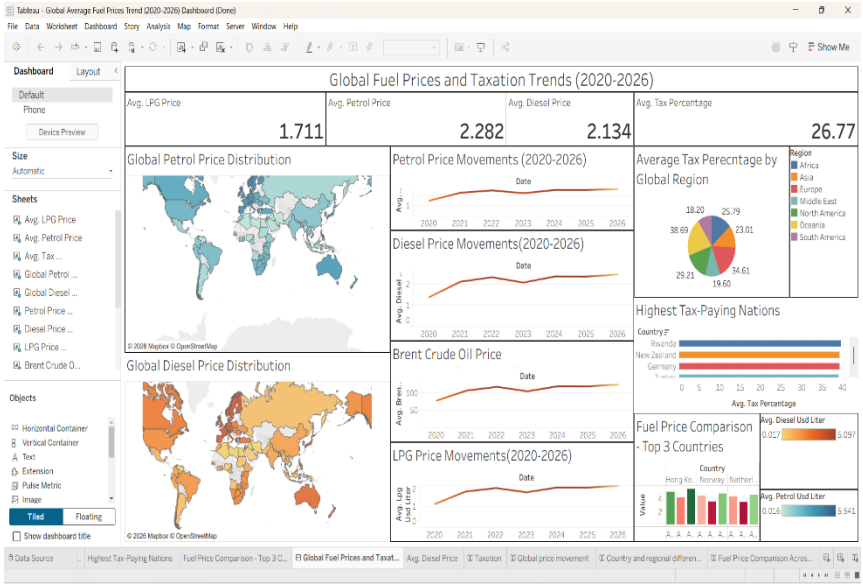
Screenshots



Website front page



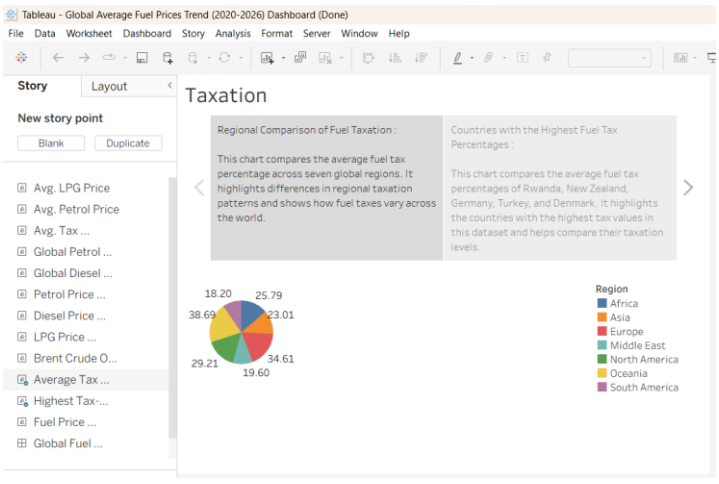
Tools and Scope



Interactive Dashboard page made from 9 different sheets from Tableau and published on tableau it is also integrated on the website

One of the Data Story created in tableau. This dashboard story provides a comprehensive overview of taxation trends from 2020 to 2026

Other Stories link provided in the document.



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8. Advantages and Disadvantages

8.1 Advantages

- Combines several fuel types and related dimensions in one analytical project.
- Makes country and time comparisons easier than inspecting raw tables alone.
- Uses Tableau dashboard interaction and Story sequencing for two different viewing needs.
- Includes taxation, subsidy, region, income and crude-oil context.
- Published Tableau links allow reviewers to access the analytical outputs online.
- A supporting Flask website provides an additional project presentation point.

8.2 Disadvantages and limitations

- The results depend on the coverage, definitions and quality of the supplied dataset.
- The dataset ends on 6 April 2026; it is not a live fuel-price feed.
- Country differences may reflect many factors not represented in the dataset.
- Visual comparison alone does not establish correlation significance or causation.
- The project does not provide forecasting or infer policy effects.
- Performance depends on the viewer's device, browser and network; measured timings must be recorded separately.

8.3 Risk and mitigation

Risk	Mitigation
Misreading a visual trend as causal	Use cautious language and state limitations.
Broken Tableau or website link	Open every link from the final PDF and demo device.
Missing screenshot evidence	Capture real outputs and place them in the report before submission.
Ambiguous units or definitions	Check axis titles, data dictionary and source documentation.
Template metadata incomplete	Add Team ID, names, dataset URL and demo URL before final upload.

9. Conclusion

The project demonstrates how a structured global fuel-price dataset can be explored through Tableau visualizations, a consolidated dashboard and a guided Story. The analysis brings together petrol, diesel and LPG prices with country, date, region, income level, subsidy level, tax percentage and Brent crude-oil values. A supporting Flask website provides a project-facing presentation layer.

The local dataset check confirms 27,468 records across 84 countries and 7 regions, with no missing values or exact duplicate rows detected. These checks support data readiness for descriptive visualization, but they do not by themselves prove that every chart, filter, published link or website feature works correctly. Those items require final functional verification.

The project's main contribution is making comparisons more accessible and organized. Its conclusions should remain descriptive, and future versions can improve source documentation, interactivity, automated refresh, accessibility and evidence-based performance reporting.

10. Future Scope

- Addadocumented, reliablesource URL and data dictionary for the dataset.
- Refresh the dataset on a defined schedule if a suitable source is available.
- Add more explicit KPI summaries and carefully chosen filters where useful.
- Add statistical analysis such as correlation with appropriate caveats, if justified by the data and project scope.
- Explore forecasting only after validating the time series and assumptions.
- Improve accessibility, mobile presentation and chart descriptions.
- Automate repeatable data-quality checks and document the test outcomes.
- Consider a database or cloud warehouse only if scale and refresh needs justify it; Snowflake was not used in this version.

11. Appendix and Submission Checklist

11.1 Dataset and source code

Dataset filename: **global_fuel_prices_Dataset.csv**. The file was available locally for the checks summarized in this report.

GitHub repository : GitHub repository

11.2 Published Tableau outputs

Dashboard: Open published Tableau Dashboard

Stories: Global Trend Story

Fuel Price comparision across Top 3 countries

Taxation

Country and regional differences

Repository: GitHub repository

11.3 Project demo video

Project demo link

11.5 Glossary

Your paragraph text

Term	Definition in this project
Dashboard	A consolidated view of multiple charts for interactive exploration.
Story	An ordered sequence of Tableau views that guides interpretation.
Tax percentage	The tax-related numeric field supplied in the dataset.
Subsidy level	A categorical field used to compare recorded subsidy groups.
Descriptive analysis	Summarizes observed data and patterns without proving causation.